A PROJECT REPORT ON

Develop application(s) to support transactions with CBDC stored on card and mobile as acquiring device

SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
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FOR THE AWARD OF THE DEGREE

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BACHELOR OF ENGINEERING (Computer Engineering)

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CERTIFICATE

This is to certify that the project report entitled

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ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on ***'Develop application(s) to support transactions with CBDC stored on card and mobile as acquiring device'****.*

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ABSTRACT

As the world is moving towards the digitalization of currencies, there is an increasing need for secure and simple payment methods that operate offline or in low infrastructure environments. This project focuses on the development of an application ecosystem that permits transactions with CBDCs resident on smart cards purchased through mobile devices. The aim is to provide a secure and interoperable architecture for P2P and merchant payments without continuous internet connectivity.

The described system consists of two main components:

- **CBDC Smart Card Wallet:** A hardware encrypted wallet where users deposit a fixed allocation of CBDC tokens.
- **Acquiring Application:** A mobile application that has the capability to start, authenticate, and complete transactions using NFC, QR Codes, or contactless methods.

Key features in the application development include offline transaction support, robust authentication and validation using cryptography, up-to-date data transmission upon reconnection, and adherence to central bank policies. The focus is on enhancing financial inclusion for remote and underserved populations who have limited access to the internet and traditional banking services.

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CHAPTER 1

INTRODUCTION

1.1 Overview

This project aims to develop a secure and accessible payment system that enables transactions using Central Bank Digital Currency (CBDC) stored on smart cards and accepted via mobile. It supports both online and offline payments, making it ideal for low-connectivity environments. The solution includes two core components: a smart card wallet to store CBDC, and an acquiring application on mobile devices to process payments via NFC or QR code. Key features include offline transaction support, secure authentication, and automatic synchronization with the central ledger when connectivity is restored. This system enhances financial inclusion and usability of CBDCs in everyday transactions, especially in underserved regions.

1.2 Motivation

The rapid development of Central Bank Digital Currencies (CBDCs) presents a unique opportunity to reshape the future of digital payments. However, realizing their full potential requires solutions that are secure, accessible, and functional even in low-connectivity environments. This project is driven by several key motivations:

- **Expand Financial Inclusion**
  - Enable digital payments for individuals in remote or underserved regions without reliable internet access.
  - Support offline transactions through smart card-based CBDC wallets.
- **Leverage Proven Technologies**
  - Use **Java Card technology** to create secure, tamper-resistant smart cards for storing and transferring CBDC.
  - Ensure all smart card communication complies with **ISO 7816** standards, promoting interoperability and security.
- **Enhance Payment Infrastructure**
  - Develop **mobile applications** for Android and iOS to serve as acquiring devices.
  - Utilize **NFC** for seamless, contactless communication between smart cards and terminals.

- **Ensure Secure Transactions**
  - Implement strong cryptographic protocols to protect CBDC payments against fraud and tampering.
  - Guarantee integrity and traceability in both online and offline modes.
- **Deliver a Seamless User Experience**
  - Focus on user-friendly interfaces for both merchants and consumers.
  - Offer fast, intuitive, and reliable payment flows, similar to existing digital wallets and contactless cards.

This motivation underpins the project's goal to deliver a practical, real-world CBDC payment solution—scalable, secure, and ready for widespread adoption.

1.3 Problem Definition and Objectives

Current CBDC systems depend heavily on internet connectivity and advanced infrastructure, limiting their usability in low-resource or offline environments. There is a need for a secure, offline-capable payment solution using smart cards and NFC-enabled mobile devices. This project aims to develop a Java Card-based system that ensures reliable, standards-compliant CBDC transactions for broader financial inclusion.

1.4 Project Scope & Limitations

Project Scope:
The project aims to design and implement a secure, token-based wallet system for Central Bank Digital Currency (CBDC), in alignment with the Reserve Bank of India's vision for digital currency. It enables end users to perform secure and real-time transactions using e (digital rupee), while maintaining strong cryptographic security and interoperability with the existing banking system. The system will:

- Facilitate CBDC wallet creation, allowing users to receive and hold digital tokens.

- Enable secure loading of funds from user bank accounts into the wallet using CBS and RTSP.
- Support CBDC redemption, allowing users to transfer funds back from their wallet to their bank account.
- Implement peer-to-peer token transfer, allowing users to send digital tokens to other users securely via QR code and cryptographic validation.
- Ensure end-to-end encryption of all token and credential data using public-private key infrastructure.

Limitations:
While the project aims to provide a robust and secure platform, there are certain limitations:

1. **Hardware Dependency**
   The system's functionality relies on access to a Secure App (smart card or secure element). This limits the use of the system on devices that lack the required hardware support.
2. **Limited Denomination Flexibility**
   Token denominations are fixed (e.g., 500), which means that the system will need to manage operations like splitting tokens or issuing change. This could limit the ability to conduct micro-transactions or result in rounding challenges during transactions.
3. **No Merchant or Third-Party App Support**
   The platform does not support integrations with third-party merchant apps, nor does it facilitate connections to open banking APIs. This limitation means that merchants must directly support the CBDC system.
4. **No Support for Offline Risk Mitigation**
   The current system design does not account for offline risk mitigation features such as double-spend protection, fraud detection, or reconciliation strategies. This limits its use in offline scenarios.
5. **Audit, Logging, and Compliance Gaps**
   At present, the system is focused solely on transaction processing, with no built-in modules for auditing, Anti-Money Laundering (AML) / Know Your Customer (KYC) checks, or regulatory compliance tracking. These features are not part of the current implementation.

1.5 Methodologies of Problem solving

1. **Requirement Analysis**
   Studied RBI's CBDC framework and identified key components: user, secure app, mobile app, RTSP, and CBS.Defined essential workflows: wallet creation, loading, redeeming, and transferring e.

2. **System Design**
   Adopted a modular, token-based architecture. Ensured secure communication and smart card compliance (ISO 7816). Modeled token lifecycle with splitting, change, and deletion.

3. **Technology Stack**
   Used Java Card for secure token storage. Developed mobile apps (Android/iOS) with NFC support. Integrated RTSP and CBS via secure APIs.

4. **Security Measures**
   Enforced end-to-end encryption using cryptographic credentials. Implemented secure token operations (lock, split, register) within the Secure App.

5. **Testing Validation**
   Performed unit and integration testing. Validated token integrity, user experience, and system interoperability.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Survey on Central Bank Digital Currency (CBDC) and Secure Wallet-based Transaction Systems

The exploration of Central Bank Digital Currency (CBDC) systems has intensified, with particular emphasis on secure and efficient offline transactions. Beer et al. introduced **PayOff**, a CBDC solution emphasizing privacy in offline payments while demonstrating the system's ability to handle high transaction volumes with limited computational resources [1]. Yang et al. proposed **DOT-M**, which utilizes Secure Elements (SE) and Trusted Execution Environments (TEE) to facilitate secure offline transactions between mobile devices, providing a robust dual offline mechanism [2]. Similarly, Androulaki et al. presented a cryptographic protocol for secure and privacy-preserving CBDC transactions using SEs, which also includes a mechanism to detect double-spending [3]. Kempen and Pouwelse explored a minimal viable CBDC implementation based on Groth-Sahai proofs, ensuring transaction privacy with selective law enforcement access under strict legal compliance [4]. Christodorescu et al. proposed a two-tier hierarchical model for offline CBDC systems to support direct, intermediary-free transactions in disconnected environments [5].

Further expanding on hardware-based approaches, Takaoğlu and Ölçer advocated for smart card-based offline payment systems, focusing on ensuring financial inclusion in areas with limited digital infrastructure [6]. Ivanov and Yan introduced **VolgaPay**, a blockchain-backed offline terminal system eliminating the need for sensitive data storage, thereby enhancing security [7]. Doğan et al. presented **ElasticPay**, a lightweight, peer-to-peer offline payment model emphasizing speed and reliability in isolated transaction settings [8]. Han et al. provided a security-focused review of offline CBDC frameworks, emphasizing authentication and double-spending prevention [9]. Additionally, Satoshi Capital Advisors provided a comprehensive classification of offline digital payment architectures, highlighting design trade-offs in custody and regulatory compliance [10].

Literature Review

1. PayOff: A Regulated Central Bank Digital Currency with Private Offline Payments

(Beer et al., 2024)

Findings: Introduces 'PayOff', a CBDC system supporting private offline payments with strong privacy guarantees and efficient performance. Demonstrates the feasibility for central banks to manage high payment loads using moderate resources.

Limitations: Offline payment message and storage requirements grow with the number of offline transactions, which may affect scalability.

Link: `https://arxiv.org/abs/2408.06956`

2. DOT-M: A Dual Offline Transaction Scheme of Central Bank Digital Currency for Trusted Mobile Devices

(Yang et al., 2022)

Findings: Proposes 'DOT-M', a scheme using Secure Elements (SE) and Trusted Execution Environments (TEE) for secure offline mobile-to-mobile CBDC transactions.

Limitations: Dependence on hardware security (SE and TEE) limits its adoption across all mobile devices.

3. Secure and Privacy-preserving CBDC Offline Payments using a Secure Element

(Androulaki et al., 2024)

Findings: Presents a privacy-preserving offline CBDC payment protocol using Secure Elements, including methods to detect and deter double-spending.

Limitations: Varies with Secure Element availability across different devices.

Link: `https://eprint.iacr.org`

4. Offline Digital Euro: A Minimum Viable CBDC using Groth-Sahai proofs

(Kempen & Pouwelse, 2024)

Findings: Introduces a lightweight CBDC model with privacy-preserving offline transactions using Groth-Sahai proofs, with lawful access capabilities.

Limitations: Implementation complexity and steep cryptographic learning curve.

Link: `https://arxiv.org/abs/2407.13776`

5. Towards a Two-Tier Hierarchical Infrastructure for CBDC Of-

fline Payments
(Christodorescu et al., 2020)
Findings: Recommends a hierarchical CBDC system enabling secure peer-to-peer transactions during network outages.
Limitations: Requires intensive collaboration between banks and intermediaries.
Link: `https://arxiv.org/abs/2012.08003`

6. Smart Card Based Offline Payment System for CBDCs
(Takaoğlu & Ölçer, 2022)
Findings: Highlights the use of smart cards to facilitate offline CBDC payments in low-connectivity areas.
Limitations: Accessibility is constrained where smart card infrastructure is lacking.
Link: `https://www.academia.edu/91253892/Smart_Card_Based_Offline_Payment_System_for_Central_Bank_Digital_Currencies`

7. System-Wide Security for Offline Payment Terminals
(Ivanov & Yan, 2021)
Findings: Proposes 'VolgaPay', a secure blockchain-enabled offline terminal system eliminating the need for sensitive data storage.
Limitations: Blockchain integration may increase computational complexity.
Link: `https://arxiv.org/abs/2107.08490`

8. ElasticPay: Instant Peer-to-Peer Offline Extended Digital Payment System
(Doğan et al., 2024)
Findings: Describes 'ElasticPay', designed for fast, secure peer-to-peer offline payments with real-world applicability.
Limitations: Large-scale resilience and security robustness still require evaluation.
Link: `https://www.mdpi.com/1424-8220/24/24/8034`

9. Review of Offline Payment Function of CBDC Considering Security Requirements
(Han et al., 2022)
Findings: A comprehensive analysis of security concerns in offline CBDC, offering architectural and protocol suggestions.
Limitations: Theoretical study—practical validation is needed.
Link: `https://www.mdpi.com/2076-3417/12/9/4488`

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10. Offline Digital Currency Payments
(Satoshi Capital Advisors, 2021)
Findings: Discusses diverse methods for offline CBDC payments, including card-based and bearer instruments.
Limitations: Addresses difficulties in ensuring transaction custody and preventing double-spending.
Link: `https://www.researchgate.net/publication/349620537_Offline_Digital_Currency_Payments`

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Assumptions and Dependencies

- **Interoperability Dependency:** The project assumes compatibility across different mobile platforms (Android), and JavaCard readers for seamless operation of the CBDC smart card system.
- **Hardware Availability:** The smart card readers, host-based emulators, NFC-enabled Android devices must be available and supported throughout the lifecycle of the application.
- **Cryptographic Dependency:** The project is dependent on the javax.crypto library and secure JavaCard APIs for encryption and secure transaction management.
- **Network Dependency:** The secure communication between the Android app, acquiring mobile device, and backend services depends on consistent internet connectivity and secure APIs.

3.2 Functional Requirements

The functional requirements define the key operations of the system.

3.2.1 System Feature 1 (CBDC Transaction Management)

- Enable load, spend, and redeem functions for the CBDC tokens using secure APIs.
- Support authentication via digital certificates, token generation, and user PIN verification.

3.2.2 System Feature 2 (Smart Card Integration)

- Perform token storage, retrieval, and deletion securely on the smart card.
- Support offline transaction verification using JavaCard applets.

3.2.3 System Feature 3 (Mobile and acquiring device Interface)

- Provide a seamless interface on Android mobile and acquiring devices for performing CBDC operations.
- Enable transaction history and wallet balance view on mobile.

3.2.4 System Feature 4 (Token Lifecycle Management)

- Handle full lifecycle of digital tokens, including issuance, transfer, and redemption.
- Implement mechanisms for rekeying and token expiry as per RBI regulations.

3.3 External Interface Requirements

3.3.1 Software Interfaces

- Integration with Retrofit for XML-based API communication.
- Integration with Docker and Maven for deployment and build automation.
- Integration with Host-Based Emulator for secure testing

3.3.2 Communication Interfaces

1. RESTful APIs: HTTP requests to retrieve csv datasets.
2. Transformers library: Uses APIs to fetch pretrained models from Hugging Face.

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- Real-time processing of transactions with response time ¡ 1s.
- System must support scaling for thousands of transactions per second.

3.4.2 Security Requirements

- Use industry-standard encryption and secure key storage (javax.crypto, JavaCard).
- Protect user data via HTTPS and authentication layers.

3.4.3 Reliability and Availability

- 99.9 percent uptime expected for backend services.
- Smart card fallback mechanisms for offline mode.

3.4.4 Usability

- User interfaces should be intuitive with minimal learning curve.
- Mobile app should require minimal configuration.

3.4.5 Maintainability

- Code must be modular and follow best practices for logging, debugging, and updating.
- Well-documented APIs and backend services.

3.5 System Requirements

3.5.1 Database Requirements

- Store metadata of transactions, wallet balance, and audit logs in PostgreSQL.
- Support data exchange in XML format over secure HTTP.

3.5.2 Software Requirements (Platform Choice)

- **Operating System and Development Environment:** Android 10+ for the mobile app; Ubuntu/Windows for backend services. Development tools include Android Studio for mobile, IntelliJ IDEA/Eclipse for backend, and Postman/Insomnia for API testing.

- **Backend Technologies:** Spring Boot with Maven for Java backend development, Hibernate for ORM, and Jackson XML/GSON for XML/JSON data parsing.

3.5.3 Hardware Requirements

- **Hardware Requirements:** Minimum Intel i5 processor with 8GB RAM and 50GB storage for backend development. ARM-based NFC-enabled Android devices with 3GB+ RAM for mobile app. JavaCard 3.0+ supported smart cards and NFC-compatible devices are essential for transactions.

3.6 Analysis Models: SDLC Model to be applied

We will follow the Agile methodology for the development of this secure CBDC smart card ecosystem. Each sprint will deliver usable features including card integration, backend security, and UI workflows. Agile allows flexibility in incorporating stakeholder feedback, adapting to security standards, and rapidly iterating on UI/UX and transaction performance.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

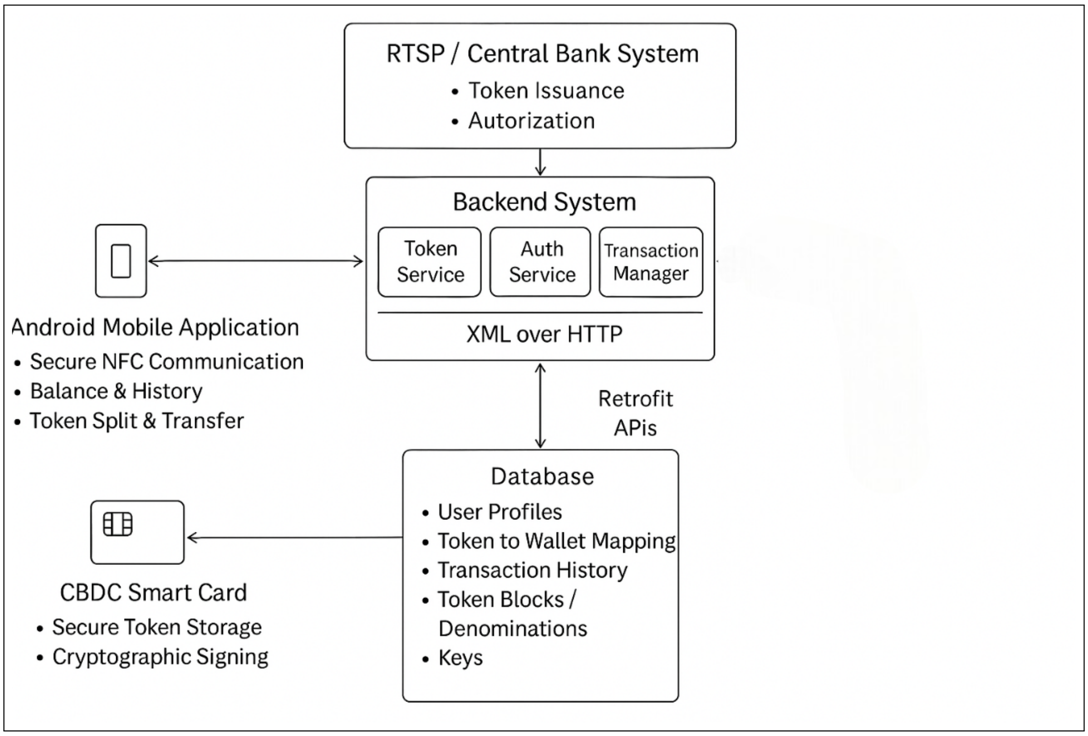


Figure 4.1: System Architecture

Working of the Application

1. User Registration & Wallet Setup

- User registers via the **e Mobile App**.
- Identity verified (OTP/KYC), key pair generated and stored in the **Secure Applet**.
- Wallet is initialized with RTSP-issued credentials.

2. Token Load (CBDC Acquisition)

- User requests to load CBDC tokens.
- Bank debits fiat via CBS, and **RTSP issues digital tokens**.

- Tokens securely loaded into the smart card via XML over HTTP.

3. Transaction at NFC Device

- User taps the card; mobile reads token info.
- Tokens are verified and split if needed.
- Secure transfer of required token value to the merchant's device.

4. Token Redemption by Merchant

- Merchant sends received tokens to RTSP.
- RTSP validates tokens and credits equivalent fiat to the merchant's account.

5. Offline Support & Storage

- App uses **SharedPreferences** for offline caching.
- Transactions sync once the device is online.

6. Monitoring & Reporting

- Backend logs all actions in **PostgreSQL** using Spring Boot + Hibernate.
- Dashboard shows wallet balance, transaction logs, and token status.

7. Deployment & Communication

- Services containerized with **Docker**, built using **Maven**.
- APIs use **XML over HTTP**, integrated with **Retrofit**.

4.2 UML - Activity Diagrams

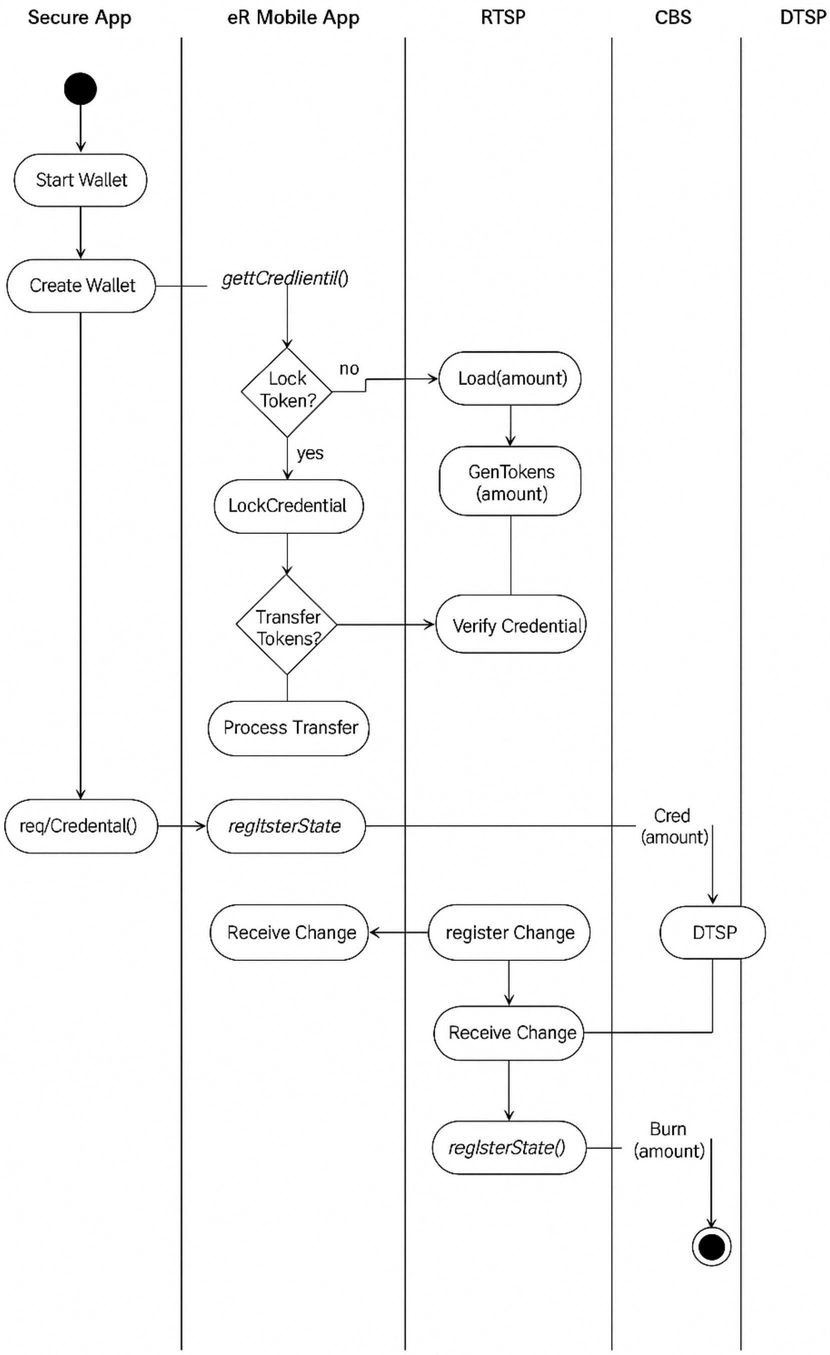


Figure 4.2: UML Activity Diagram

4.3 UML Usecase Diagram

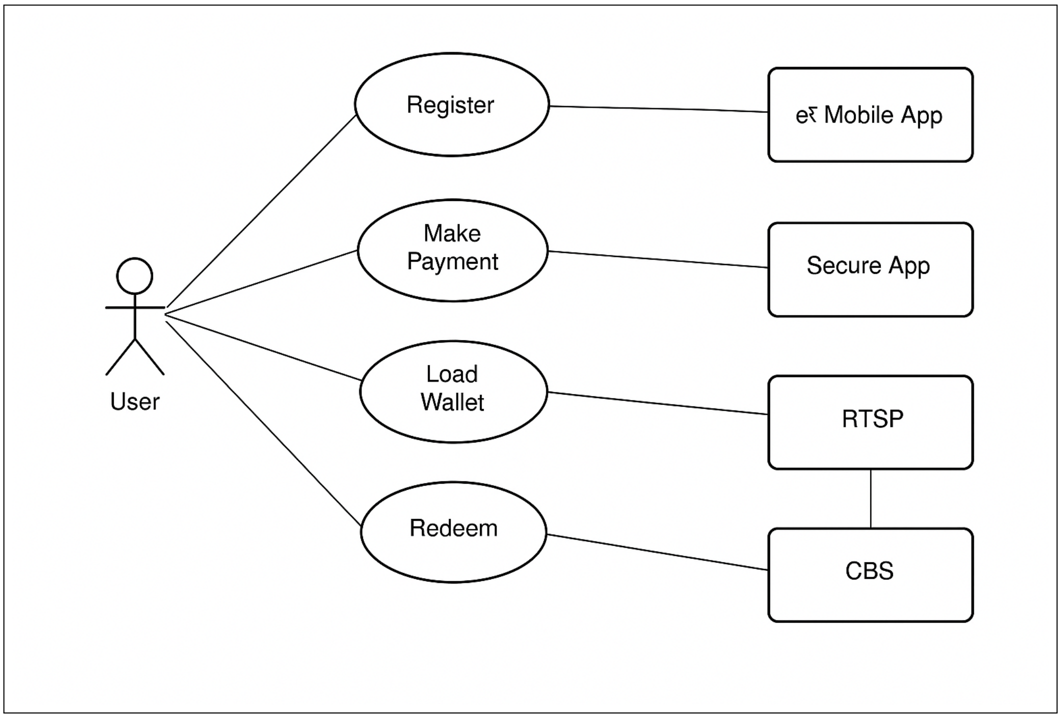


Figure 4.3: Use case diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Elements

The reconciled estimates bring together the development timeframe, resource allocation, and budget to create a comprehensive project overview:

- **Development Hours:**This timeframe considers the complexity of the efficiency enhancement techniques implementation and potential regulatory compliance challenges. .
- **Resource Allocation** $0 — A dedicated team of team members with specialized roles will be allocated, A dedicated team of team members with specialized roles will be allocated, along with an external domain expert to ensure compliance along with an external domain expert to ensure compliance.
- **Budget Allocation :** The budget estimate has been allocated based on detailed resource costs, software licenses, and potential legal and compliance consulting.

5.1.2 Project Resources

A dedicated team of specialized team members will be allocated, along with an external domain expert. Additionally, hardware devices will be required for the further steps of the project.

5.2 Risk Management

To ensure secure and efficient deployment of the CBDC-based smart card and mobile acquiring system, proactive risk identification, monitoring, and mitigation strategies must be in place. The following section outlines potential risks and corresponding mitigation strategies for the project.

5.2.1 Risk Identification

- **Technical Risk:** Integrating NFC-based secure communication with Java Card, mobile app may lead to unforeseen technical bottlenecks or interoperability issues.

- **Security Risk:** As the system deals with monetary value and offline transactions, any breach in security or flaw in token handling can lead to double-spending or fraud.
- **Regulatory/Compliance Risk:** The CBDC ecosystem must comply with national financial regulations and digital currency guidelines, which are evolving.
- **Resource Risk:** Hardware such as Java Cards, secure element and backend infrastructure might face availability or cost constraints.
- **Integration Risk:** Ensuring smooth communication between Secure App, RTSP, CBS, and mobile apps is complex and may lead to delays or inconsistent behavior.
- **Schedule Risk:** As multiple components and stakeholders are involved, synchronizing development and testing schedules may be challenging.

5.2.2 Risk Analysis

- **Computational Limitations:** Medium likelihood, high impact. Insufficient resources could drastically slow model training and reduce experimentation capabilities.
- **Model Accuracy and Efficiency Trade-offs:** High likelihood, medium impact. Difficulty in maintaining accuracy while improving efficiency could affect the overall success of the project.
- **Data Handling and Availability:** Medium likelihood, medium impact. Poor-quality data or insufficient volumes could result in ineffective model training, impacting both efficiency and accuracy.
- **Integration and Scalability Issues:** Medium likelihood, medium impact. Delays in integrating and scaling models could slow deployment and affect system performance.
- **Software or Technical Failures:** Low likelihood, medium impact. These failures could cause temporary delays but are generally manageable with proper planning and backups.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- **Risk Avoidance: :** Early planning, detailed requirement analysis, and use of industry standards will help avoid many foreseeable risks.
- **Risk Mitigation :** For unavoidable risks, mitigation strategies such as security audits, fallback mechanisms, and contingency planning will be implemented.
- **Risk Transfer :** Some responsibilities like backend security may be outsourced to vendors or certified service providers.
- **Risk Acceptance :** Low-impact risks, such as minor delays or performance trade-offs in non-critical modules, will be accepted with a monitoring plan.

5.3 Project Schedule

5.3.1 Project Schedule

Project Plan for Developing a CBDC Card-Based Transaction System with Mobile Acquiring Devices

- **Phase 1: Requirements Gathering and System Design**
  - Analyze existing CBDC solutions and standards (e.g., ISO 7816, RBI's CBDC architecture).
  - Identify stakeholder requirements (users, banks, regulatory authorities).
  - Define system components: Secure App (Java Card), e Mobile App, RTSP backend, CBS integration.
  - Finalize technical architecture including cryptography, offline capability, and secure communication channels.
  - Prepare a detailed project plan outlining deliverables, timelines, and resources.
- **Phase 2: Module Development**
  - **Secure App (Java Card):** Develop applet for wallet storage, token management, key exchange, and encryption/decryption.

  - **Mobile Application:** Build Android/iOS app for registration, wallet load/redeem, transaction initiation, and user interface.
  - **Acquiring Device Interface:** Implement secure NFC-based token reading and transfer capability.
  - **RTSP Backend:** Develop services for token issuance, trust setup, wallet updates, audit logging, and compliance.
  - **CBS Integration:** Integrate APIs for real-time fiat debit/credit and wallet synchronization.

- **Phase 3: Integration and Testing**
  - Perform module-level unit testing.
  - Conduct system integration testing for workflows like registration, wallet load, spend, and redeem.
  - Test across various devices (mobile online/offline and Java Cards).
  - Evaluate system performance (load testing, latency, token validation time).
  - Conduct security testing including penetration tests and encryption robustness.

- **Phase 4: Deployment and Maintenance**
  - Deploy system in a sandboxed environment for pilot runs with limited users.
  - Monitor logs, performance statistics, and user feedback.
  - Fix bugs and optimize based on pilot feedback.
  - Release updates with security patches, enhancements, and compliance updates.
  - Plan for future features like biometric verification, QR support, and multi-currency token handling.

5.3.2 Timeline:

- Phase 1: Requirements Gathering and Planning – **1 Month**
- Phase 2: Software Development – **6 Months**
- Phase 3: Software Testing – **1 Month**
- Phase 4: Software Maintenance – **1 Month**

Total Duration: Approximately 9 Months

5.4 Team Organization

5.4.1 Team structure

- Team Lead
- Team of 4 members
- Industrial expert
- Domain Expert

5.4.2 Management reporting and communication

- Regular team meetings for project coordination.
- Bi-weekly reporting to management on resource utilization and budget allocation.
- Communication through collaboration tools and regular status reports to ensure transparency and accountability.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

6.1.1 Secure App

This module resides on a secure Java Card (or equivalent secure element) and is responsible for securely storing CBDC tokens. It handles critical operations such as:

- Key generation and exchange
- Token encryption/decryption
- Token creation, update, and deletion
- Offline authentication and trust establishment with mobile devices
- Preventing double-spending via token lifecycle management

6.1.2 e Mobile App

This mobile application acts as the user-facing interface for initiating various CBDC operations. Key functionalities include:

- Device registration and secure key handshake
- Wallet loading and redemption
- Transaction initiation and offline support
- User interface to display token balance and transaction history

6.1.3 Acquiring Device (Mobile App) Interface

This module acts as an acquiring device during retail transactions. It is responsible for:

- Initiating and processing transactions using NFC with the Secure App
- Validating token authenticity offline
- Syncing transaction logs with the RTSP backend when online
- Ensuring seamless user experience for merchant-side CBDC acceptance

6.2 Algorithm Details

1. **Initialize System Components**
   - Power on and initialize the Secure App, e Mobile App, RTSP backend, and acquiring device.
2. **Registration and Mutual Trust Setup**
   - Generate and exchange cryptographic keys between Secure App, Mobile App, and RTSP.
   - Establish mutual trust using secure token-based authentication (SA1, SA2).
   - Store verified identities and keys on respective devices.
3. **Wallet Load Operation**
   - User initiates “Load Wallet” via the e Mobile App.
   - Secure App creates and encrypts a credential block with token info using RTSP’s public key.
   - RTSP debits the user’s bank account via CBS and credits the pool account.
   - RTSP generates updated tokens, encrypts them with the Secure App’s public key, and sends them back.
   - Secure App decrypts and securely stores the tokens.
4. **Offline/Online Transaction at Acquiring Device**
   - User taps the CBDC card on the merchant’s acquiring device (NFC-based).
   - Acquiring device verifies token validity and balance offline.
   - Transaction amount is deducted from Secure App tokens.
   - Acquiring device logs the transaction for syncing when back online.
5. **Wallet Redemption Operation**
   - User initiates “Redeem” in the mobile app.
   - Secure App encrypts the redemption token block and sends it to RTSP.

- RTSP decrypts and locks corresponding tokens.
- RTSP credits the user's account via CBS.
- Secure App deletes redeemed tokens to avoid double-spending.

6. **Token Validation and Sync**
   - RTSP periodically validates token integrity and consistency across devices.
   - Acquiring device syncs logged transactions with RTSP for reconciliation.

7. **Security and Monitoring**
   - Constant monitoring of token lifecycle and trust tokens (SA1/SA2).
   - All sensitive operations are cryptographically verified.

6.3 Tools and Technologies Used

Category	Technologies
Backend	Spring Boot, Java
Database	PostgreSQL, JPA (Hibernate)
Communication / APIs	XML over HTTP, Retrofit
DevOps / Deployment	Maven, Docker
Android Application	Kotlin

Table 6.1: Technologies Used in the Project

CHAPTER 7

SOFTWARE TESTING

7.1 Testing

1. Functional Testing: Verify that the proposed sentence selection algorithm performs as expected by accurately ranking and selecting sentences.
2. Performance Testing: Measure the computational efficiency of the model, including memory usage and inference time, when processing long documents.
3. Cross-Domain Adaptation Testing: Evaluate the model's ability to adapt from SHC training data to long document classification (LDC) tasks.
4. Regression Testing: Ensure that modifications to the sentence selection algorithm or the model do not negatively impact overall classification performance.

7.2 Test cases & Test Results

7.2.1 Load Amount

Request JSON


```
{
 "amount": 25,
 "txnId": "kpIuKkbdNquItDLGsAzVNibghHl04MCoPxz",
 "userToken": {
  "tokenId": "QeOXxSRfDU22N1VOMzDY4rmUidi6jJyH3qA",
  "denomination": 75
 }
}
```


Response JSON


```
{
 "amount": 25,
 "txnId": "kpIuKkbdNquItDLGsAzVNibghHl04MCoPxz",
 "addToken": {
  "tokenId": "gVnKpLzQwXyTeRmJuCiBdOzHaLs83NqWpMt",
  "denomination": 100
 },
```

```
 "deleteToken": {
  "tokenId": "QeOXxSRfDU22N1VOMzDY4rmUidi6jJyH3qA",
  "denomination": 75
 }
}
```

7.2.2 Redeem Amount

Request JSON

```
{
 "amount": 18,
 "txnId": "x6EoJxwZw6Xw9KBNrjoaOlwClNqt3Uyc1Ex",
 "userToken": {
  "tokenId": "ouWBbIzvv10726yoJbr4nfEwXPRoEod5LBq",
  "denomination": 86
 }
}
```

Response JSON

```
{
 "amount": 18,
 "txnId": "x6EoJxwZw6Xw9KBNrjoaOlwClNqt3Uyc1Ex",
 "addToken": {
  "tokenId": "JzRmQwTkXpLuNsVyAeBgChDoFiLx29MvKtE",
  "denomination": 68
 }
 "deleteToken": {
  "tokenId": "ouWBbIzvv10726yoJbr4nfEwXPRoEod5LBq",
  "denomination": 86
 }
}
```

7.2.3 Send Money

Request JSON

```
{
 "amount": 15,
 "txnId": "aLtYpZxWmNcVoRiKeDjBuHsFgMqT47LyEz",
 "userToken": {
```

```
  "tokenId": "QwErTyUiOpAsDfGhJkLzXcVbNm91PoRtYu",
  "denomination": 73
 }
}
```


Response JSON


```
{
 "amount": 15,
 "txnId": "x6EoJxwZw6Xw9KBNrjoaOlwClNqt3Uyc1Ex",
 "addToken": {
  "tokenId": "MnBvCxZaSdFgJhKlQwErTyUiOp87LkJmNh",
  "denomination": 88
 }
 "deleteToken": {
  "tokenId": "QwErTyUiOpAsDfGhJkLzXcVbNm91PoRtYu",
  "denomination": 73
 }
}
```


7.2.4 Sync Balance

Request JSON


```
{
 "tokenId": "XpLwMqTeNzVkRyAsDfGhJuIoCb93LmKtEy",
 "denomination": 44
}
```


Response JSON


```
{
 "addToken": {
  "tokenId": "ZaQsXwEdCvFrTgByNhUjMiLoPk67JmReTu",
  "denomination": 60
 }
 "deleteToken": {
  "tokenId": "QwErTyUiOpAsDfGhJkLzXcVbNm91PoRtYu",
  "denomination": 44
 }
}
```

CHAPTER 8

RESULTS

8.1 Screenshots

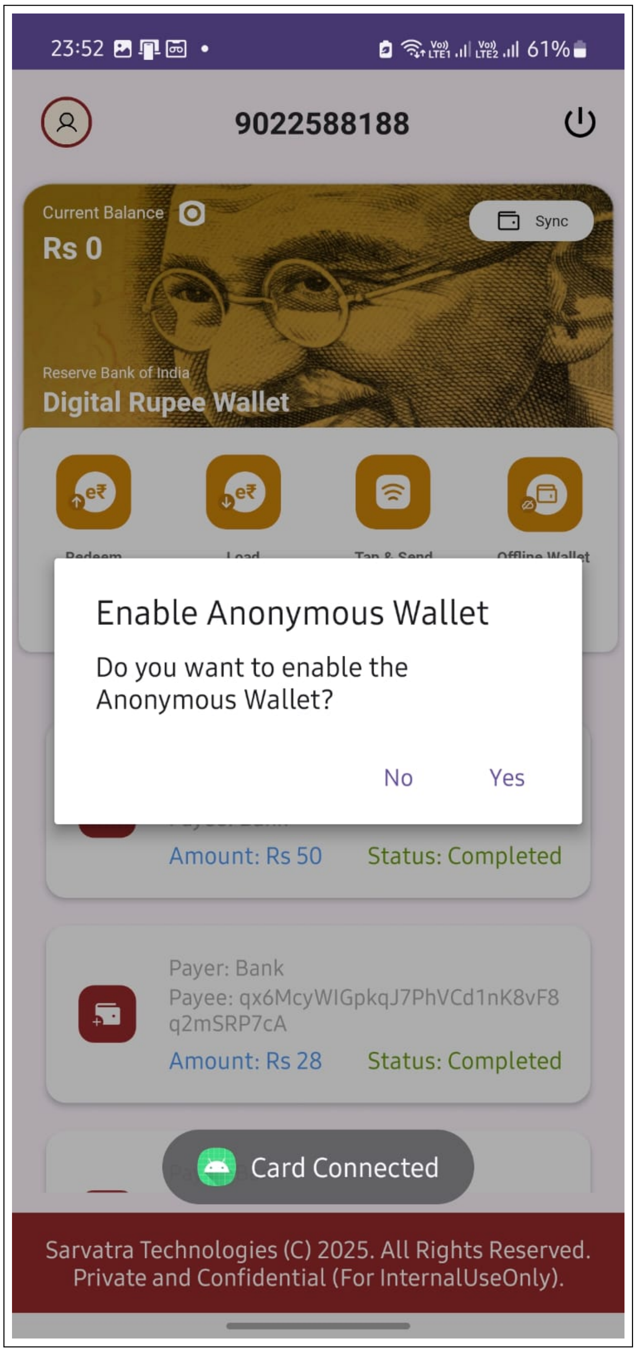


Figure 8.1: Registration of Wallet

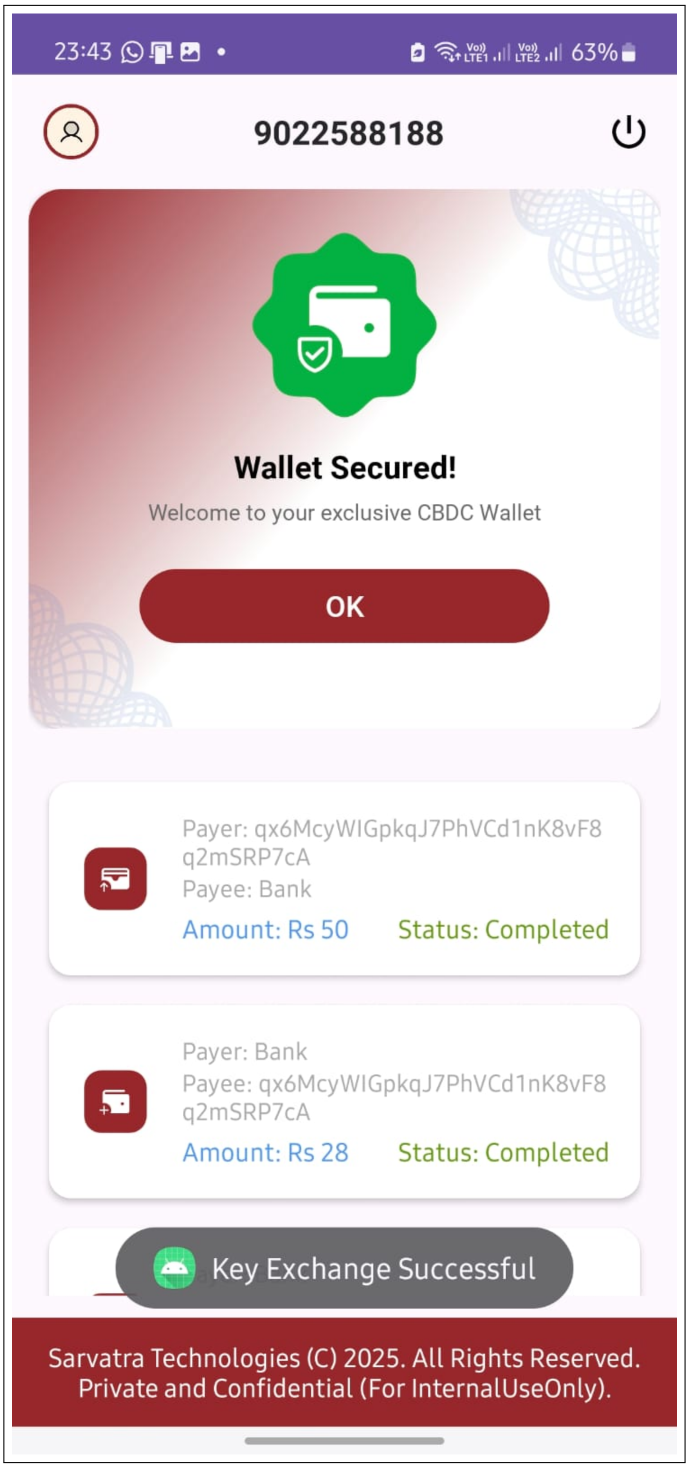


Figure 8.2: Registration successfully completed image

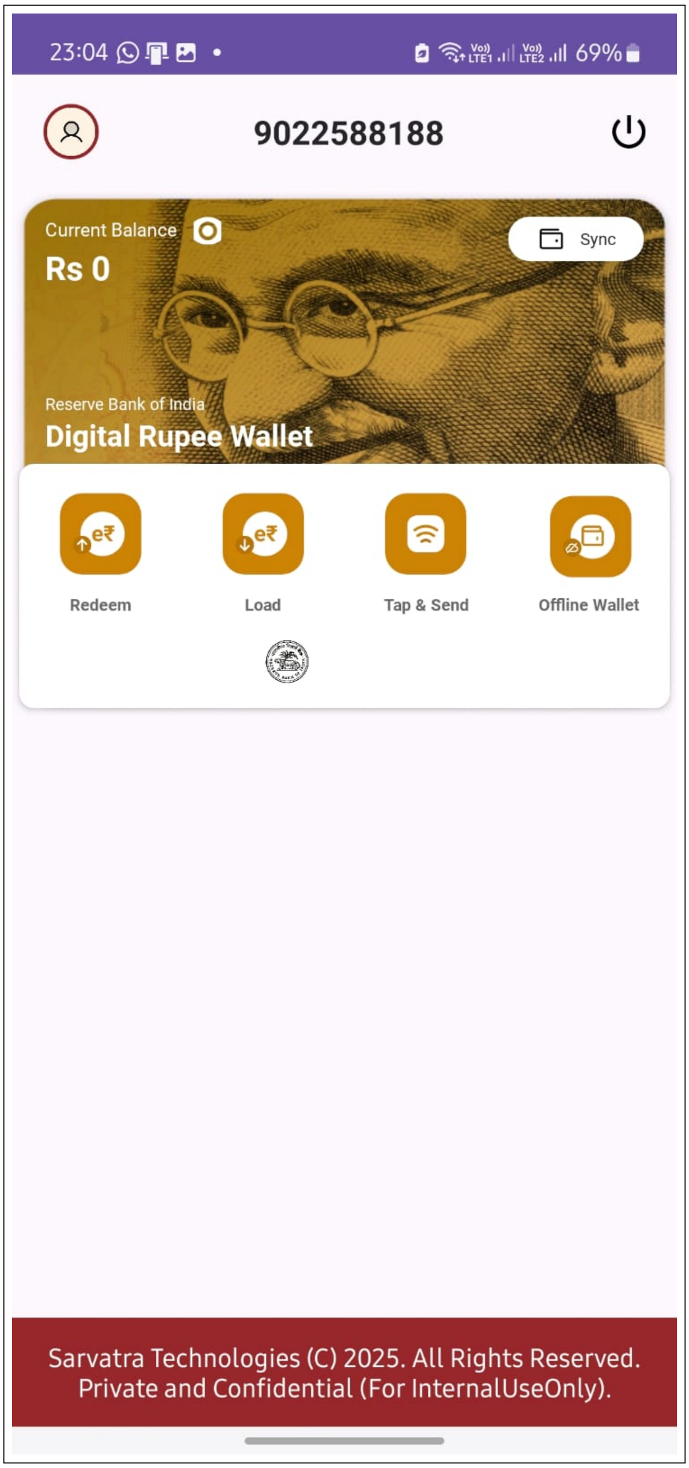


Figure 8.3: Home Page

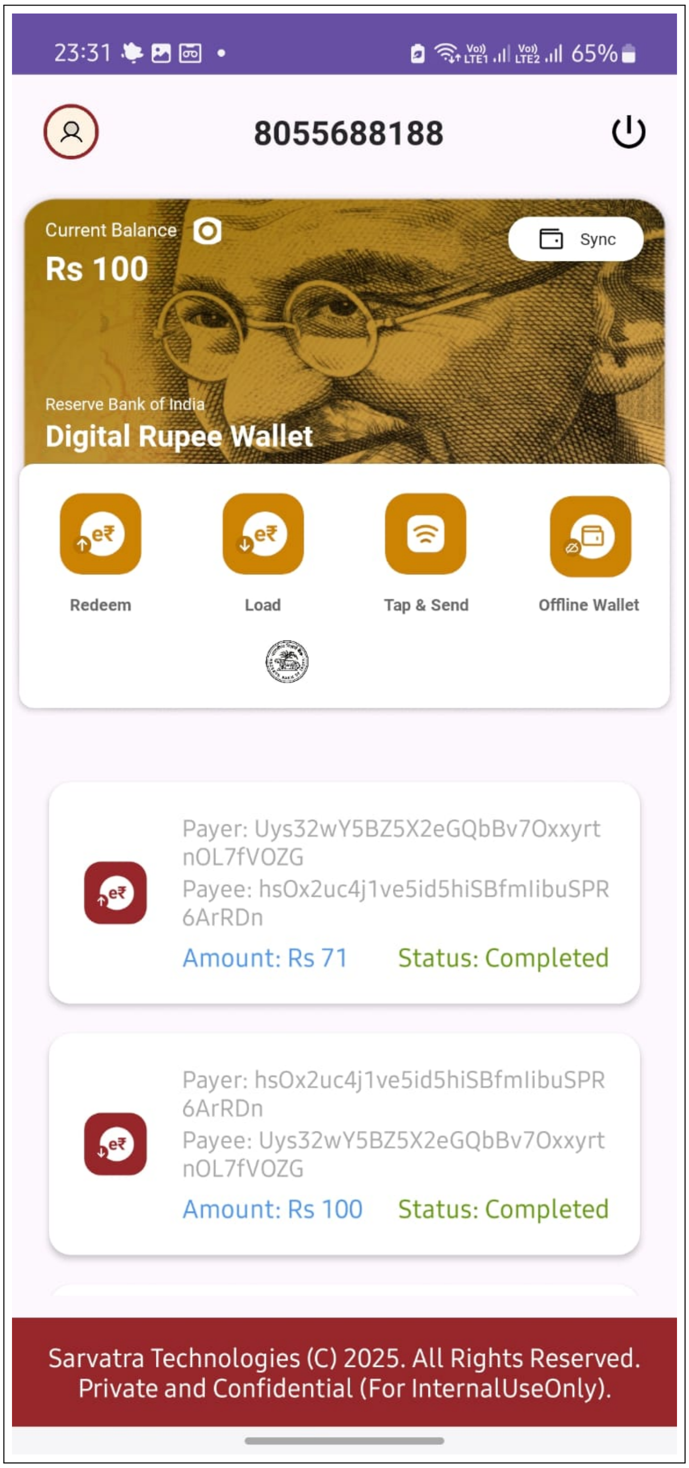


Figure 8.4: Home page with transactions

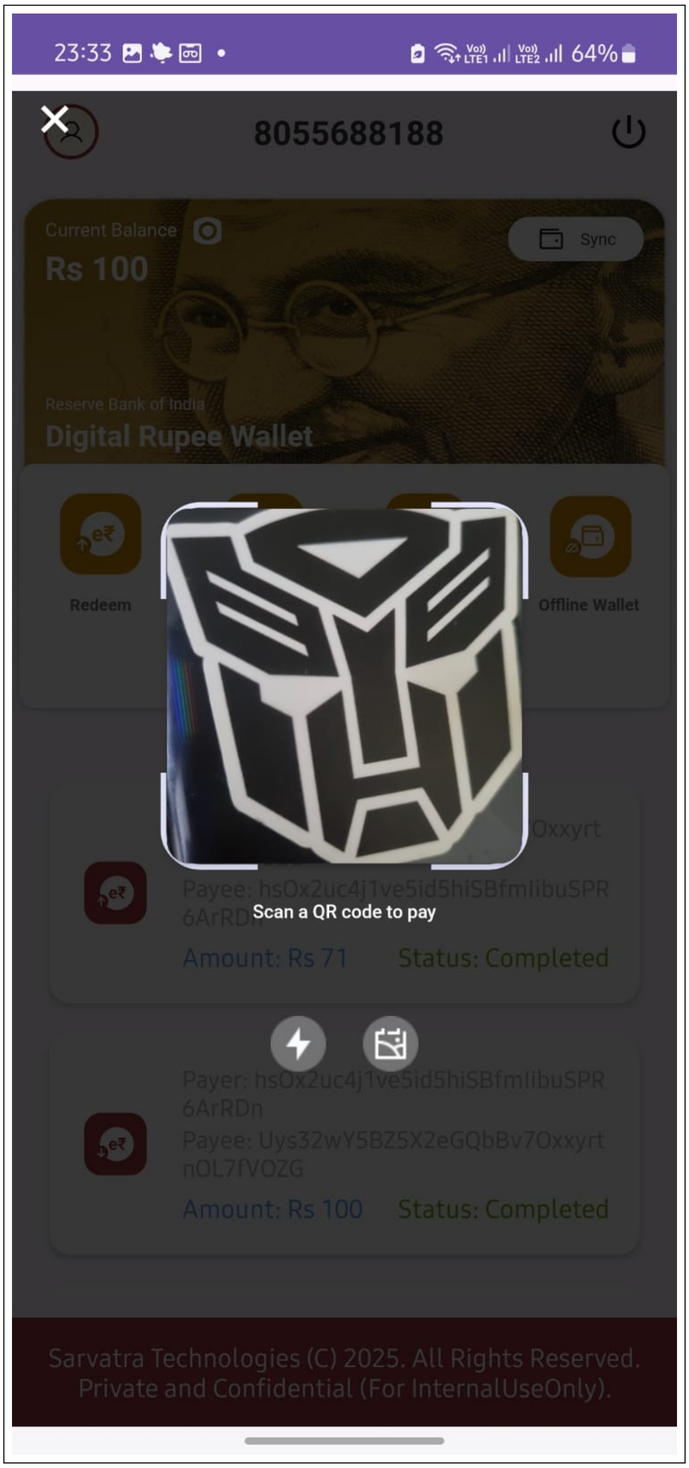


Figure 8.5: Qr Scanner for sending tokens

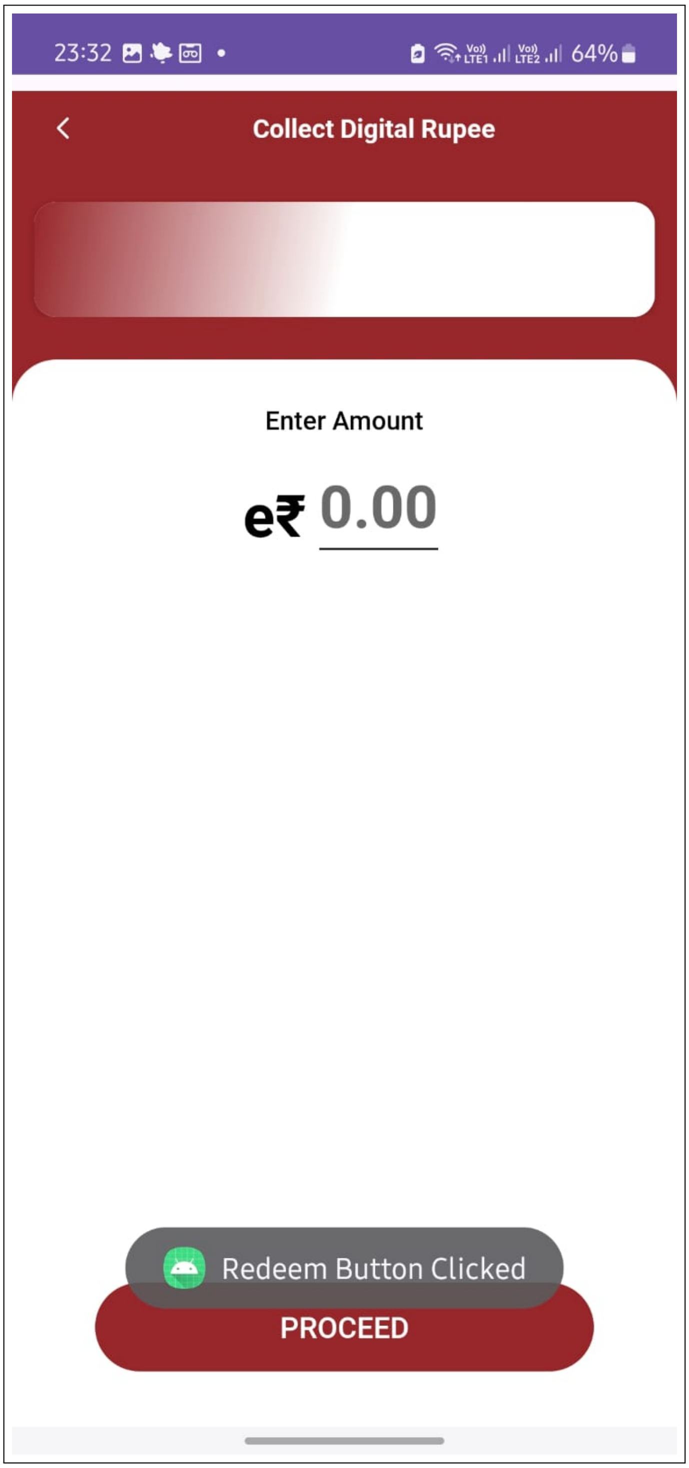


Figure 8.6: Image for entering amount while load, redeem and send

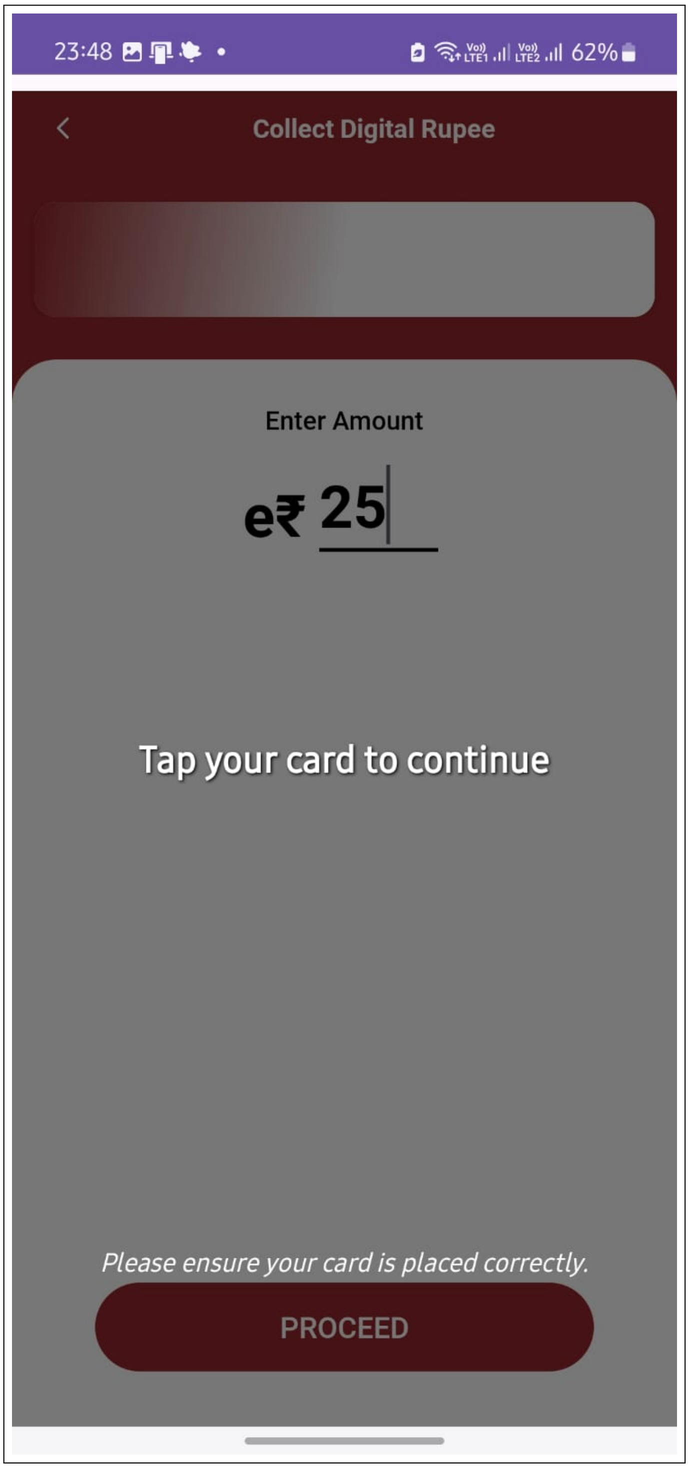


Figure 8.7: Image for completing all transaction(load, redeem and send)

CHAPTER 9
CONCLUSIONS

9.1 Conclusion

This project successfully demonstrates the development of an offline-capable, secure digital currency transaction system using CBDC (Central Bank Digital Currency) stored on a secure card and transacted via mobile devices. By leveraging mutual trust establishment, public-key cryptography, and token-based mechanisms, the system ensures privacy, security, and offline usability—addressing real-world constraints in rural and infrastructure-poor areas. The modular workflow comprising wallet load, redemption, and offline transactions has been designed and tested for practical feasibility. This solution aligns with the growing demand for sovereign digital currencies and secure payment infrastructure at national scale.

9.2 Future Work

1. **Interoperability with Other CBDC Networks :**
    - Extend the system for cross-border or inter-bank CBDC interoperability, integrating with international regulatory standards.
2. **Biometric or Multi-Factor Authentication**
    - Incorporate fingerprint or facial recognition-based authentication on POS/mobile apps for enhanced security.
3. **Scalability Token Refresh**
    - Introduce token refresh mechanisms for time-bound token validity and scalability for millions of users.
4. **AI-Driven Fraud Detection**
    - Integrate machine learning models to detect abnormal spending patterns and potential misuse.
5. **Support for Additional Devices**
    - Expand support for feature phones using USSD/SMS-based interfaces for greater financial inclusion.
6. **CBDC Programmability**
    - Enable programmable CBDC features for conditional transfers (e.g., subsidies, refunds, etc.).

9.3 Applications

1. **Offline Payments in Remote Areas**
    - Facilitates digital transactions in regions with limited or no internet connectivity.
2. **Retail Merchant Transactions**
    - Acts as a secure alternative to physical cash for everyday purchases at POS terminals.
3. **Government Subsidy Distribution**
    - Can be used for direct benefit transfers to beneficiaries without requiring a constant internet connection.
4. **Disaster Relief Emergency Payments**
    fast deployment of aid in crisis zones where infrastructure is down.
5. **Transportation Toll Booths**
    - Supports quick, tap-based digital payments at transit points without latency.
6. **Military Defense**
    - Allows secure and trackable payments in field operations with no reliance on real-time network access.

CHAPTER 10
APPENDIX

10.1 Appendix A: Computational and Mathematical Modeling in CBDC Smart Card System

The development of a secure CBDC (Central Bank Digital Currency) Smart Card ecosystem involves multiple complex components such as secure key exchange, token issuance, real-time transaction processing, and hardware-level communication. This makes the problem multidimensional and rich for analysis from the perspective of feasibility, computational complexity, NP classifications, and mathematical modeling. Below is a breakdown of these components:

1. Satisfiability Analysis

Satisfiability analysis, typically used in the domain of SAT (Boolean satisfiability) problems, involves determining if a set of logical constraints can be satisfied simultaneously. In the context of CBDC smart card operations, satisfiability analysis could be loosely applied when validating cryptographic conditions (e.g., token validity, mutual trust, and policy-based access control). However, since our system does not reduce directly to a Boolean satisfiability problem, satisfiability analysis has limited direct applicability, except in protocol verification or rule-based compliance checks.

2. Computational Complexity

The project includes various operations such as real-time digital token validation, RSA encryption, and secure APDU (Application Protocol Data Unit) communication, all of which contribute to the overall computational complexity. While not NP-hard in a traditional sense, this problem is computationally demanding due to the following factors:

- **Cryptographic Operations:** Encryption and decryption (e.g., RSA, AES) are computationally intensive, particularly on constrained smart card hardware. Efficient modular exponentiation is crucial here.
- **Real-Time Constraints:** Operations must complete within strict time limits to ensure user experience (especially in POS systems) is not degraded.

- **Secure Communication:** Managing secure APDU exchanges and ensuring proper handshake/acknowledgement patterns between the Secure App, RTSP, and Mobile App adds both communication and verification overhead.
- **Memory and I/O Constraints:** Smart cards have limited memory and computation capacity, which increases the complexity of fitting secure protocols efficiently.

3. P-Type Problems

Some aspects of the system, such as formatting XML messages, generating UUIDs, or validating token formats, are P-type problems, i.e., solvable in polynomial time using standard algorithms. These routine tasks contrast sharply with the computational load imposed by cryptographic operations and secure multi-party coordination. Hence, the overall system is a hybrid, combining P-type subproblems with more intensive cryptographic and synchronization tasks.

4. Mathematical Models

Several mathematical concepts and models underpin the design and implementation of the system:

- **Cryptographic Mathematics:** The system employs modular arithmetic and number theory concepts used in RSA encryption, digital signatures, and key exchange protocols.
- **Graph Models for Transaction Flow:** Secure token transfers and validations across multiple systems (Mobile App $\leftrightarrow$ Secure App $\leftrightarrow$ RTSP $\leftrightarrow$ CBS) can be modeled using directed graphs with weighted edges representing computational or time cost.
- **Finite-State Machines (FSMs):** Smart cards and host emulators use FSMs to model stateful APDU communication and protocol flows, ensuring all operations occur in the correct sequence.
- **Queuing Models and Resource Allocation:** Used to optimize transaction throughput in real-time systems like POS terminals, especially when dealing with concurrent requests.
- **Verification & Formal Methods:** Mathematical logic and formal languages can be used to model and verify security properties, such as non-repudiation, data integrity, and mutual authentication.

Develop application(s) to support transactions with CBDC stored on card and mobile as acquiring device

Conclusion

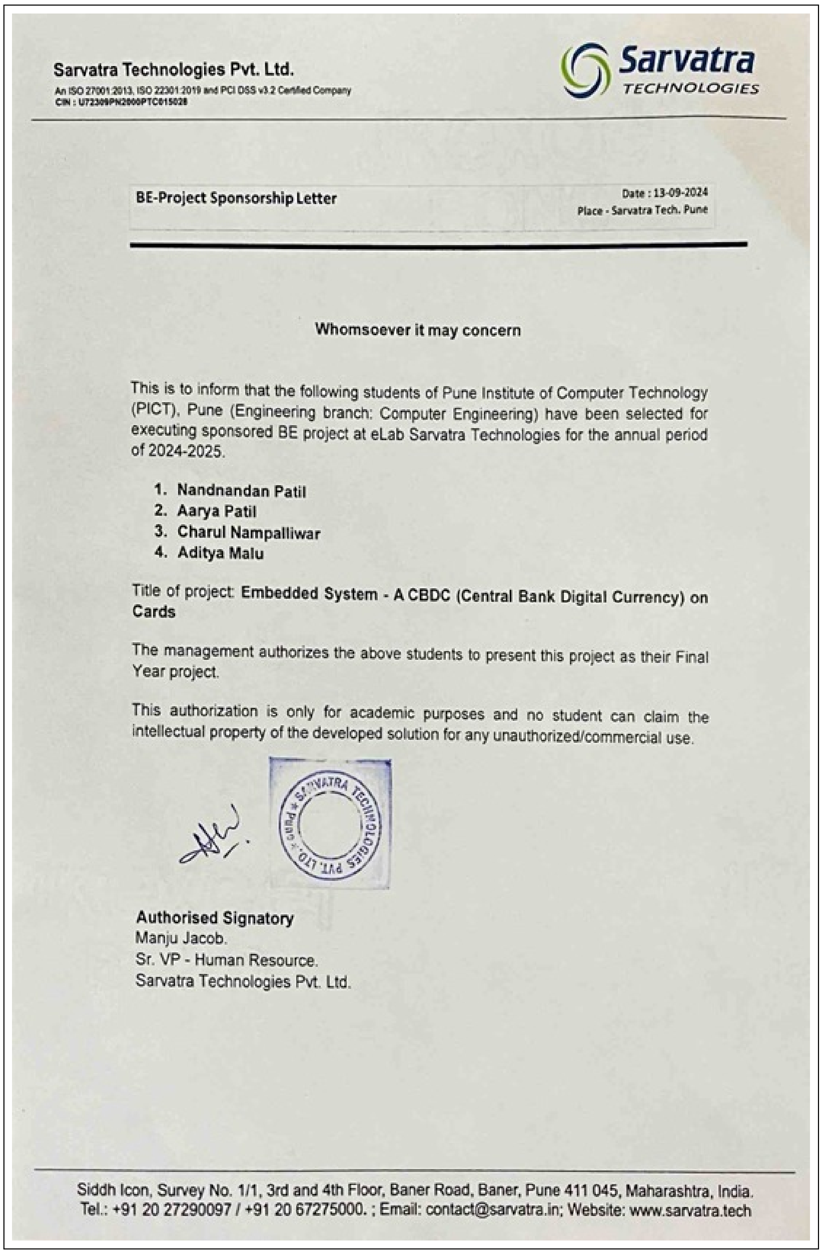
Although not directly mappable to NP-hard or NP-complete problems, the CBDC Smart Card ecosystem presents a computationally intensive challenge, particularly in the cryptographic and hardware-level protocol design domains. While routine tasks remain solvable in polynomial time, the system's feasibility is highly dependent on resource optimization, protocol correctness, and real-time hardware constraints. Leveraging cryptographic mathematics, distributed system principles, and FSM-based design, the system achieves its goals of secure, efficient, and scalable digital currency operations.

10.2 Appendix B

- **International Conference on Sustainability, Innovation & Technology (ICSIT 2025):** *Context-Aware Resource Sharing in a Multi Application Smart Card Emulator* (Submitted)
- **2025 IEEE Madhya Pradesh Section Conference (MPCON):** *Context-Aware Resource Sharing in a Multi Application Smart Card Emulator* (Submitted)
- **2025 2nd International Conference on Computing and Data Science (ICCDS):** *Context-Aware Resource Sharing in a Multi Application Smart Card Emulator* (Submitted)
- **2024 IEEE 16th International Conference on Computational Intelligence and Communication Networks (CICN):** *Comparative Study of Functionality Isolation in Multi-Application Smart Cards* (Submitted and Published)

10.3 Appendix C

Sponsorship Letter



Sarvatra Technologies Pvt. Ltd.
An ISO 27001:2013, ISO 22301:2019 and PCI DSS v3.2 Certified Company
CIN : U72309PN2000PTC015028

Sarvatra TECHNOLOGIES

BE-Project Sponsorship Letter

Date : 13-09-2024
Place - Sarvatra Tech. Pune

Whomsoever it may concern

This is to inform that the following students of Pune Institute of Computer Technology (PICT), Pune (Engineering branch: Computer Engineering) have been selected for executing sponsored BE project at eLab Sarvatra Technologies for the annual period of 2024-2025.

1. **Nandnandan Patil**
2. **Aarya Patil**
3. **Charul Nampalliwar**
4. **Aditya Malu**

Title of project: **Embedded System - A CBDC (Central Bank Digital Currency) on Cards**

The management authorizes the above students to present this project as their Final Year project.

This authorization is only for academic purposes and no student can claim the intellectual property of the developed solution for any unauthorized/commercial use.

SARVATRA TECHNOLOGIES PVT. LTD. * Pune *

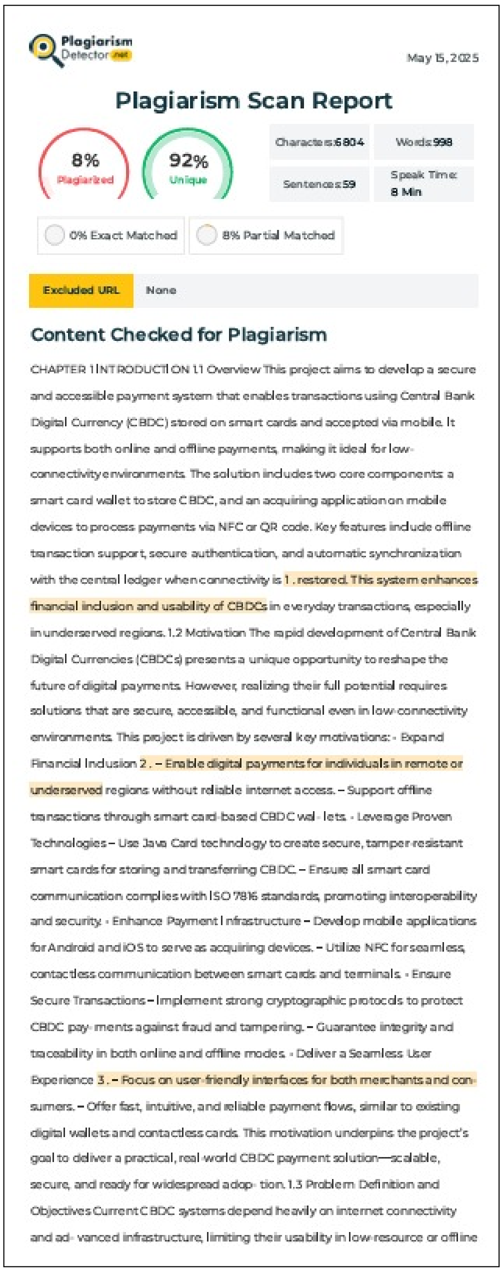
Authorised Signatory
Manju Jacob.
Sr. VP - Human Resource.
Sarvatra Technologies Pvt. Ltd.

Siddh Icon, Survey No. 1/1, 3rd and 4th Floor, Baner Road, Baner, Pune 411 045, Maharashtra, India.
Tel.: +91 20 27290097 / +91 20 67275000. ; Email: contact@sarvatra.in; Website: www.sarvatra.tech

Figure 10.1: Plagiarism Report

10.4 Appendix D

Plagiarism Report of project report.


Plagiarism Detector .net

May 15, 2025

Plagiarism Scan Report

8% Plagiarized

92% Unique

Characters: 6804	Words: 998
Sentences: 59	Speak Time: 8 Min

0% Exact Matched

8% Partial Matched

Excluded URL	None

Content Checked for Plagiarism

CHAPTER 1 INTRODUCTI ON 1.1 Overview This project aims to develop a secure and accessible payment system that enables transactions using Central Bank Digital Currency (CBDC) stored on smart cards and accepted via mobile. It supports both online and offline payments, making it ideal for low-connectivity environments. The solution includes two core components: a smart card wallet to store CBDC, and an acquiring application on mobile devices to process payments via NFC or QR code. Key features include offline transaction support, secure authentication, and automatic synchronization with the central ledger when connectivity is 1 . restored. This system enhances financial inclusion and usability of CBDCs in everyday transactions, especially in underserved regions. 1.2 Motivation The rapid development of Central Bank Digital Currencies (CBDCs) presents a unique opportunity to reshape the future of digital payments. However, realizing their full potential requires solutions that are secure, accessible, and functional even in low-connectivity environments. This project is driven by several key motivations: • Expand Financial Inclusion 2 . – Enable digital payments for individuals in remote or underserved regions without reliable internet access. – Support offline transactions through smart card-based CBDC wal- lets. • Leverage Proven Technologies – Use Java Card technology to create secure, tamper-resistant smart cards for storing and transferring CBDC. – Ensure all smart card communication complies with ISO 7816 standards, promoting interoperability and security. • Enhance Payment I nfrastructure – Develop mobile applications for Android and iOS to serve as acquiring devices. – Utilize NFC for seamless, contactless communication between smart cards and terminals. • Ensure Secure Transactions – Implement strong cryptographic protocols to protect CBDC pay- ments against fraud and tampering. – Guarantee integrity and traceability in both online and offline modes. • Deliver a Seamless User Experience 3 . – Focus on user-friendly interfaces for both merchants and con- sumers. – Offer fast, intuitive, and reliable payment flows, similar to existing digital wallets and contactless cards. This motivation underpins the project's goal to deliver a practical, real-world CBDC payment solution—scalable, secure, and ready for widespread adop- tion. 1.3 Problem Definition and Objectives Current CBDC systems depend heavily on internet connectivity and ad- vanced infrastructure, limiting their usability in low-resource or offline

Figure 10.2: Plagiarism Report

CHAPTER 11
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LIST OF ABBREVIATIONS

Abbreviation	Illustration
APDU	Application Protocol Data Unit
RTSP	Reserve Token Service Provider
JWT	JSON Web Token
CBS	Core Banking System
AID	Application Identifier
RSA	Rivest-Shamir-Adleman
NFC	Near Field Communication
CBDC	Central Bank Digital Currency
XML	eXtensible Markup Language
JPA	Java Persistence API
UUID	Universally Unique Identifier
UI	User Interface

HTTP	HyperText Transfer Protocol
GSON	Google's Serialization Library for JSON

LIST OF FIGURES

LIST OF TABLES